

1 B mass of electron negligible compared to nucleons

question similar to asking for nuclear density:

$$\rho_{\text{material}} \approx \frac{\text{mass}}{V_{\text{nucleus}}} = \frac{\cancel{4}u}{\frac{\cancel{4}}{\cancel{3}}\pi\cancel{r_{\text{nucleus}}}^3}$$
$$\approx \frac{10^{-27}}{10^{-45}} = 10^{18} \text{ kg m}^{-3}$$

2 A B $\propto (Et^2)^{0.2}$